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9 May 1978
SUPERSEDING
MIL-D-43278
28 September 1964

MILITARY SPECIFICATION

DISPENSER, GRAPHITE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable dispenser containing graphite.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

SS-G-659	- Graphite, Dry (Lubricating).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

FSC 4930

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The graphite dispenser shall consist of a tube (body), nozzle, and a cap. A typical dispenser is shown in Figure 1.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Graphite.

3.3.1.1 Composition. Graphite shall be in accordance with SS-G-659.

3.3.1.2 Particle size. The graphite shall be of such particle size as to enable not less than 88 percent of the granules to pass through a U.S. Series No. 200 standard sieve when tested as specified in 4.5.2.1.

3.4 Design. The tube shall store a minimum of one-fifth ounce of graphite and allow sufficient air space for normal operation of the dispenser. The dispenser shall be designed to dispense graphite from the tube and through an opening in the nozzle by merely squeezing the tube. The cap shall screw or snap onto the nozzle.

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3.5 Construction.

3.5.1 Tube. The tube shall be fabricated from a flexible plastic material. One end of the tube shall be closed and the other end of the tube shall have the nozzle joined to it to form a leakproof assembly.

3.5.2 Nozzle. The nozzle shall be fabricated from a rigid (hard) plastic material. The nozzle shall be threaded or beaded externally to accomodate the cap. The outside diameter of the nozzle shall be tapered with the outside diameter of the smallest end (outlet), a maximum of 0.125-inch (3.175 mm). The discharge opening of the nozzle shall be not more than 0.047-inch (1.191 mm) diameter plus or minus 0.005-inch (.127 mm).

3.5.3 Cap. The cap shall be fabricated from a rigid or semirigid plastic material. The cap shall be threaded or grooved internally. The outside surface at one end of the cap may be provided with flutes or grooves for ease of assembly and disassembly with the nozzle. The inside surface of the cap (excluding thread surface) shall be tapered similar to the taper on the nozzle so that the tapered surfaces will contact each other when the cap fully engages the nozzle.

3.6 Environmental conditions.

3.6.1 Operating temperature. Unless otherwise specified (see 6.2), the dispensers, each containing one-fifth ounce of graphite, shall after storage for 48 hours at plus 160° F, plus or minus 5° F (plus 71° C, plus or minus 3° C), and minus 45° F, plus or minus 5° F (minus 43° C, plus or minus 3° C), be capable of normal operation in a temperature range between minus 25° F, plus or minus 5° F (minus 32° C, plus or minus 3° C) and plus 125° F, plus or minus 5° F (plus 52° C, plus or minus 3° C) without cracking, leaking, failure to return to normal shape, fracture, or damage of any kind. The dispensers shall be tested as specified in 4.5.2.2.

3.7 Finish. The completed dispenser shall be clean and smooth with no flash, rough edges, gouges, dents, lumps, or discolorations.

3.8 Identification marking. Each graphite dispenser shall be identified in accordance with MIL-STD-130.

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3.9 Workmanship. The dispenser shall conform to the quality and grade of the product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.3.1 Sample size. Five dispensers shall be subjected to first article inspection.

4.3.2 Inspection routine. The samples shall be subjected to the inspections specified in 4.5.1 and 4.5.2.

4.3.3 Failures. Failure of one or more units shall be cause for refusal to grant first article approval.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The dispenser shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Design not as specified.
103. Construction not as specified.
104. Environmental conditions not as specified.
105. Finish not as specified.
106. Workmanship not as specified.

Minor

201. Identification marking missing or illegible.

4.5.2 Tests.

4.5.2.1 Sieve test. Using a mechanical shaker capable of vibrating a U.S. Series No. 200 standard testing sieve 900 to 1,000 times a minute for not less than 45 minutes and producing 2 impacts each motor revolution, sift 2 ounces (0.57 grams) of graphite through the sieve. Weigh sifted graphite and residue on sieve and calculate the percentage of graphite passed through sieve. Passage of less than 88 percent shall constitute failure of this test.

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4.5.2.2 Performance. To determine compliance with 3.6.1 the dispensers containing one-fifth ounce of graphite shall be capable of dispensing 80 percent of the contents of each dispenser by normal operation (squeezed) under each of the following conditions:

At plus 125° F, plus or minus 5° F, (plus 52° C, plus or minus 3° C).
At minus 25° F, plus or minus 5° F, (minus 32° C, plus or minus 3° C).

Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
108.	Quantities not as specified for Level A or Level B.
109.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Twelve dispensers shall be contained in a close-fitting box conforming to PPP-B-566, Variety 1, style optional; PPP-B-676, Variety 1, type, class, and style optional, or in the contractors standard commercial container.

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5.1.1.1 Consolidation. Twelve boxes of dispensers, prepared as specified in 5.1.1, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be closed and sealed as specified for Method V in the appendix to the box specification.

5.1.2 Commercial. Dispensers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Dispensers, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Dispensers, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional in quantities not to exceed the weight or size limitations of the box. Box closure shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Dispensers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The graphite dispensers are used to lubricate outdoor equipment, especially in extremes of climate where liquid lubricant does not suffice.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required for inspection and approval (see 3.2, 4.3, and 6.3).
(c) Operating temperatures, when different (see 3.6.1).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of five units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.4 Information figure. Figure 1 shows a type of dispenser which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another dispenser which conforms to this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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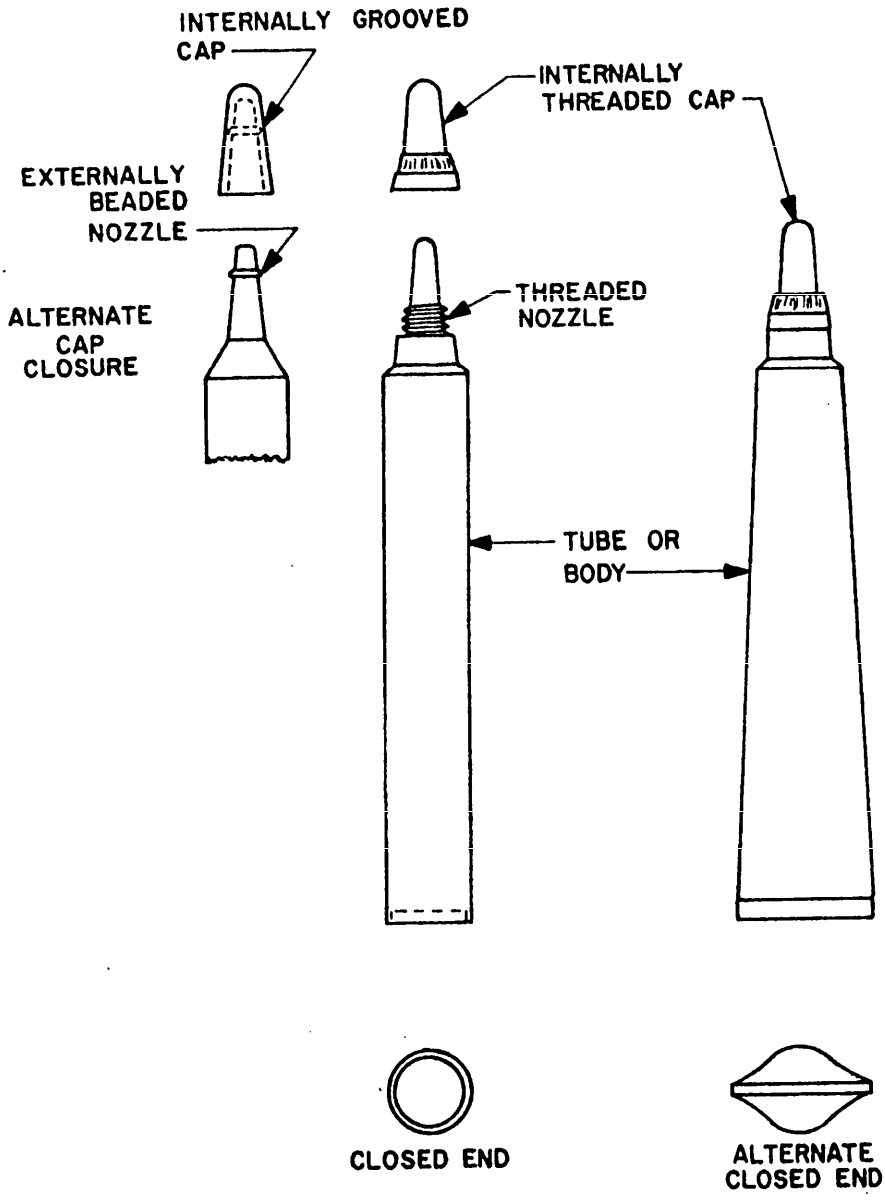


FIGURE I. Dispenser, graphite; 1/5-ounce capacity.

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-43278A Dispenser, Graphite

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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